Class Test 2 **Date: November 11, 2024**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

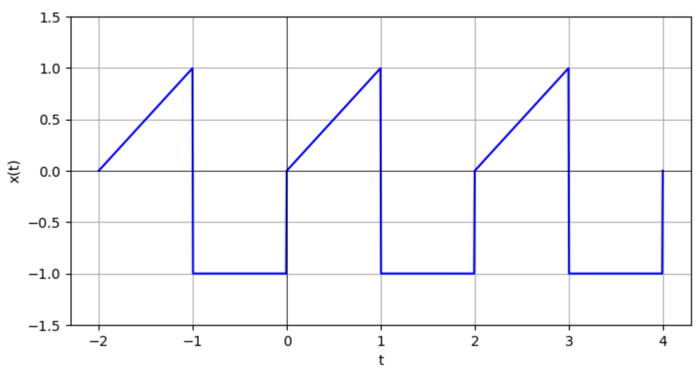
Department of Computer Science and Engineering

L-2/T-2 CSE 219: Signals and Linear Systems

Time: 25 minutes **Marks: 20**

Student Name: ______________________________ Student No: ____________

1. (i) The graph below shows a periodic signal $x(t)$, where x-axis of the graph denotes time t and y-axis denotes $x(t)$. Determine the complex form of the Fourier series representation of $x(t)$. You don't need to plot the spectrum graph. (8 Marks)



(ii) Consider fourier series coefficient b_k for a periodic signal, where (4 Marks)

$$b_k = \begin{cases} 0, & k = 0 \\ (j)^k \cdot \frac{\sin\left(\frac{k\pi}{4}\right)}{k\pi}, & \text{otherwise} \end{cases}$$

Use appropriate fourier series property to determine whether the signal is real signal or not.

2. Compute the convolution $y[n] = x[n] * h[n]$ for the following system (assume $\alpha \neq \beta$ and $0 < \alpha, \beta \leq 1$). Determine whether this system satisfies these properties: memorylessness, stability and causality.(8 Marks)

$$x[n] = \alpha^n u[n]$$
$$h[n] = \beta^n u[n]$$